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United States Patent Application Publication

20250279782

Kind Code

A1

Publication Date

September 04, 2025

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FEEDBACK DIVIDER IN A PLL USED AS A PHASE/FREQUENCY DETECTOR

Abstract

A phase-locked loop (PLL) uses a feedback divider to provide phase and frequency information for faster lock. The PLL includes a time to digital converter that generates first phase information indicative of a phase difference between a reference clock and a feedback clock. A digital filter receives the first phase information and generates an oscillator control signal used by an oscillator to generate an oscillator output. A feedback divider uses a ripple counter to divide the oscillator output to generate the feedback clock. One sample of the ripple counter provides second phase information and two samples of the ripple counter provide frequency information, which are used to generate the oscillator control signal. The one sample of the ripple counter is compared to an expected value when the PLL is locked to generate second phase information and a difference between the two samples is used to generate the frequency information.

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Family ID: 96880573

Appl. No.: 18/592910

Filed: March 01, 2024

Publication Classification

Int. Cl.: H03L7/099 (20060101); H03L7/083 (20060101); H03L7/093 (20060101)

U.S. Cl.:

CPC H03L7/0992 (20130101); H03L7/083 (20130101); H03L7/093 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application relates to application Ser. No. _____, having attorney docket number 026-0483, naming Sheng Jue Peh as inventor, filed the same day as the present application, and entitled “PLL Using A DCO Period Length Feedback Clock Pulse To Determine Phase Error and TDC Gain”, which application is incorporated herein by reference in its entirety.

BACKGROUND

Field of the Invention

[0002] This application relates to phase-locked loops (PLLs) and more particularly to frequency and phase detection using a feedback divider.

Description of the Related Art

[0003] In digital PLLs, the digital phase detector can have limited range leading to slow locking or even failure to lock. In such PLLs fast lock techniques are typically used to ensure that the PLL eventually locks and does so in a reasonable amount of time. FIG. 1 illustrates a conceptual block diagram of one technique to achieve fast lock. The PLL **100** is a conventional all digital PLL with a time-to-digital converter (TDC) **102** providing a digital phase difference **104** between the reference clock signal (fref) **106** and the feedback clock (fb_clk) signal **108** to the digital loop filter **110**. When locked, the feedback clock signal **108** is aligned in phase and frequency to the reference clock signal **106**. The digital loop filter **110** supplies a control signal **112** to the digitally controlled oscillator (DCO)/voltage controlled oscillator (VCO). The DCO/VCO **114** generates a DCO_OUT signal **116** that is supplied to the feedback divider **118**, which in turn generates the feedback clock **108**. In addition, the DCO_OUT signal **116** is supplied to the accumulator **120**. The accumulator also receives the fref signal and the accumulator counts the number of DCO_OUT cycles that occur in a reference clock signal. The number of DCO_OUT cycles indicates the frequency of the DCO_OUT signal since the reference clock frequency is known. That number (or a value based on that number) is supplied to the digital loop filter **110**, which adjusts the oscillator control signal **112** based on the accumulator value. Normally, when in lock or close to lock, the TDC phase error information is sufficient to allow the PLL to remain in lock. The use of the accumulator avoids the problem of the limited range of the TDC resulting in small adjustments being made to the oscillator control signal potentially leading to a long time to lock or even failure to lock. The need for the accumulator **120** results in more circuitry that is required to run on the high speed DCO_OUT signal and increased power consumption.

SUMMARY OF EMBODIMENTS OF THE INVENTION

[0004] Accordingly, embodiments combine the functionality of the accumulator with the feedback divider thereby avoiding the need for extra circuitry for the accumulator. In an embodiment, a phase-locked loop (PLL) includes a phase detector circuit coupled to receive a reference clock signal and a feedback clock signal and generate first phase error information indicative of a phase difference between the reference clock signal and the feedback clock signal. A digital filter is coupled to receive the first phase error information and generate a filter output using the first phase error information, the filter output being used to generate an oscillator control signal. An oscillator is configured to generate an oscillator output signal based, at least in part, on the oscillator control signal. A feedback divider is coupled to the oscillator output signal and generates the feedback clock signal. The feedback divider is sampled to supply frequency error information and second phase error information used to adjust the oscillator control signal.

[0005] In another embodiment a method includes generating first phase error information indicative of a phase difference between the reference clock signal and a feedback clock signal in a phase-locked loop (PLL). A loop filter receives the first phase error information and generates an

oscillator control signal using, at least in part, the first phase error information. An oscillator receives the oscillator control signal and generates an oscillator output signal. The method further includes generating a feedback divider signal in a feedback divider coupled to the oscillator output signal and sampling the feedback divider to generate frequency error information and second phase error information. The method further includes generating the oscillator control based, at least in part, on at least one of the frequency error information and the second phase error information. [0006] In another embodiment a phase-locked loop (PLL) includes a time to digital converter circuit coupled to receive a reference clock signal and a feedback clock signal and generate first phase error information indicative of a phase difference between the reference clock signal and the feedback clock signal. A digital filter is coupled to receive the first phase error information and generate a filter output using the first phase error information, the filter output being used to generate an oscillator control signal. An oscillator generates an oscillator output signal based, at least in part, on the oscillator control signal. A feedback divider circuit includes a ripple counter coupled to the oscillator output signal and the feedback divider is configured to generate the feedback clock signal. Samples of the ripple counter are used to generate frequency error information and to generate second phase error information and at least one of the second phase error information and the frequency error information is used to generate the oscillator control signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present invention may be better understood, and its numerous objects, features, and advantages made apparent to those skilled in the art by referencing the accompanying drawings.

[0008] FIG. 1 illustrates a conceptual block diagram of a technique to achieve fast lock using an accumulator running on the oscillator output.

[0009] FIG. 2 illustrates a high level block diagram of an embodiment that uses the feedback divider with a ripple counter to provide coarse phase and frequency information to help achieve a faster lock in the PLL.

[0010] FIG. 3 illustrates an embodiment of a ripple counter.

[0011] FIG. 4A illustrates a high level block diagram of an embodiment that uses the feedback divider with a ripple counter to provide coarse phase and frequency information to help achieve a faster lock in the PLL.

[0012] FIG. 4B illustrates how frequency and phase error is determined in an embodiment.

[0013] FIG. 5 illustrates a timing diagram showing how the count value of the ripple counter provides frequency and phase information that can be used to adjust the frequency of the DCO.

[0014] FIG. 6 illustrates an embodiment of feedback clock logic used to provide the rising and falling edges of the feedback clock signal.

[0015] FIG. 7 is a timing diagram showing how the frequency and phase information provided by the ripple counter does not affect the loop filter when the PLL is locked.

[0016] The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION

[0017] FIG. 2 illustrates a high level diagram of an embodiment of a PLL that combines the functionality of the accumulator with the feedback divider thereby avoiding the need to have extra circuitry to provide a quicker lock. In addition, the phase/frequency error is scaled nicely to the output frequency, which can make the response of the PLL fully linear. In FIG. 2, the PLL 200 includes a time-to-digital converter (TDC) 202 providing the phase difference 204 between the reference clock signal (fref) 206 and the feedback clock (fb_clk) signal 208 to the digital loop filter

210. The digital loop filter **210** supplies a control signal **212** to the oscillator (DCO/VCO) **214** to control the frequency of the oscillator output signal (DCO_OUT) **216**. The DCO/VCO can be implemented in many different ways as is well known to those of skill in the art. In an embodiment the DCO/VCO includes a digital to analog converter to convert the digital signal from the digital loop filter to an analog value to control a conventional LC tank oscillator. In other embodiments, a varactor array is digitally controlled to adjust a frequency of the LC oscillator. Various combinations of digital and analog techniques can be used to tune the oscillator depending on its implementation. In still other embodiments the digital control signal **212** controls a ring oscillator through analog and/or digital mechanisms. For ease of reference, the oscillator **214** will be referred to as DCO **214** regardless of its particular implementation and the oscillator output signal is referred herein as DCO_OUT. The DCO **214** generates the DCO_OUT signal **216** that is supplied to the feedback divider **218**, which in turn generates the feedback signal **208**. In an embodiment the feedback divider **218** includes a ripple counter. The feedback divider count value provides phase and frequency information used to adjust the control signal **212** as explained further herein. The ripple counter can be implemented as a down counter or an up counter.

[0018] FIG. **3** illustrates an embodiment of a ripple counter **300** included in feedback divider **222**. The ripple counter **300** has a maximum count value of 32 (0 to 31) but can be reset to a desired value when the count value reaches a desired count to achieve a flexible divide value. For example, the ripple counter can be reset when the count value reaches 12 (or any other desired number) assuming the ripple counter is a down counter. Alternatively, the ripple counter can be loaded with a desired count value, e.g., 23 and count down to zero. When the ripple counter is an up counter and starts from 0, the ripple counter is reset when the counter reaches a desired value, e.g., 20 (or any other count desired). Thus, the divide value of the ripple counter is programmable in many different ways as is known to those of skill in the art. Programmability is useful to achieve flexibility in the frequency generated by the oscillator **114** and to provide the ability to implement a fractional-N PLL where the divide value is changed to achieve a desired non-integer frequency. Note that q_0 is the least significant bit in the implementation of the ripple counter shown in FIG. **3**. [0019] FIG. **4A**, **4B**, and FIG. **5** illustrate how sampled counter values provide phase and frequency information that can be used to adjust the DCO **214** to more quickly lock the DCO to the desired phase and frequency. FIG. **4A** includes the same elements as the PLL shown in FIG. **2** and in addition shows separately control logic **402** whose use is explained in conjunction with FIG. **5**. FIG. **5** illustrates various clock signals present in the system of FIG. **4A** including the reference clock signal (fref) **206**, a retimed fref signal (fref_re_n) **415**, which in an embodiment is two DCO_OUT cycles after fref **206** and inverted, the feedback clock (fb_clk) signal **208** and DCO_OUT clock **216**. Retiming the fref signal (fref_re_n) **415** ensures synchronization between fref **206** and the DCO_OUT signal **216**. As can be seen in FIG. **5**, the reference clock signal **206** and the feedback clock signal **208** are not aligned meaning the PLL is not locked. The control logic samples the ripple counter output (div_cnt) **410**, i.e., ($q_0 < 0 > : q_4 >$), using the retimed fref signal. [0020] In the embodiment illustrated in FIG. **5** the divider utilizes the ripple counter as a down counter which counts from 23 to 0 to provide a divide by 24 to generate the feedback clock signal. Note that the feedback clock signal (fb_clk) rising edge occurs responsive to the count value reaching 0 and the falling edge occurs responsive to the count value reaching 12 to provide a 50% duty cycle. Of course, the duty cycle of the feedback clock signal can be adjusted by changing the value of the ripple counter causing the falling edge of the feedback clock. Of course, the rising edge can also be changed by changing to a different count value to cause the rising edge. The edges of the feedback clock signal can be determined according to the count value and SR logic shown in FIG. **6**.

[0021] While FIG. **5** illustrates an embodiment in which the ripple counter is implemented as a down counter in other embodiments the ripple counter is an up counter counting from 0 to 23 with the reset being asserted when the count value equals 23 (or another desired count value). In still

other embodiments the ripple counter counts down from a maximum count value, e.g., 31, to a desired count value, or up from a desired initial count value to the maximum count value. Other approaches to the ripple counter are possible and known to those of skill in the art.

[0022] Still referring to FIGS. 4A and 5, the control logic **402** samples the outputs **410** (div_cnt) of the ripple counter at **502** on the rising edge of the retimed fref_n signal. The control logic samples a count value of “1” at **502**. The feedback clock rising edge occurs at **504** responsive to the ripple counter counting down to 0. The ripple counter is then reloaded with the desired count value, e.g., 23 and then the ripple counter starts counting down. Loading the ripple counter with the desired value before counting down is well known in the art. For example, preset and clear functionality on D flip-flops can be used to load the desired start value. In one or more embodiments each of the flip-flops is configured to load either a 1 or 0 on assertion of reset according to another signal not shown in FIG. 4A. Other approaches can of course be used according to the particular implementation of the ripple counter. The feedback clock signal falling edge occurs at **506**. The control logic **402** then samples the ripple counter on the next rising edge of the retimed fref_n signal at **508**. At that time the control logic samples a count value of “2”. The difference between the current count value “2” and the previous count value “1” provides the frequency information that can be used to adjust the control signal **212** supplied to the DCO **214**. Here because the difference is +1, that indicates the DCO_OUT clock signal is too fast. If the difference were negative, that would indicate the DCO_OUT clock signal is too slow. That frequency information allows the control signal **212** supplied to the DCO **214** to be adjusted by an amount **403** corresponding to the frequency error indicated by the difference between the successive measurements. The frequency error **403** is a constant multiplied by the difference between successive and thus is correlated directly with the DCO output. Thus, a difference of +2 results in an error correction twice that of +1. The constant is stored, e.g., in non-volatile memory and retrieved by control logic **402**.

[0023] Phase error can be determined from the frequency error that is determined using the output **410** of the ripple counter. To measure the frequency error, embodiments measure the difference between the current and previous values. In Z transform notation, as shown in FIG. 4B, the frequency error **433** is determined as $(1 - z.\text{sup.} - 1)$ in block **432**. Summing up the frequency error successively in summer **434** results in the phase error $(1/(1 - z.\text{sup.} - 1))$. The phase error **436** can then be utilized by the digital loop filter **210** in a manner similar to phase error **204** supplied by the TDC. Referring to FIG. 4A, in an embodiment the control logic **402** causes the selector circuit **420** to select the phase error **436** determined from the ripple counter in place of the output **204** from the TDC **202** in situations in which faster lock is desired. When the phase error **436** settles back to zero, the control logic causes the selector circuit to switch back to the TDC output **204**.

[0024] In an embodiment the frequency error from FIG. 4B may be used alone or with phase error **436** to adjust the control signal **212** provided to the DCO **214**. Both frequency error **433** and phase error **436** are multiplied by a gain that is equivalent to the number of TDC delays per DCO period. In an embodiment in which the frequency error **433** is used to help the DCO **214** lock faster, the frequency error **433** is summed with the output of the DCO in summer **404** to provide an integrator function.

[0025] The loop filter **210** in various embodiments provides the desired combination of proportional, integral, and derivative control based on the needs of the particular system. Adjusting the output of the loop filter based on the phase error **436** and/or the frequency error **433** helps reduce the lock time as the error can be more quickly addressed than by waiting for the phase/frequency error information supplied by the TDC to be used to attempt to lock. The appropriate place to incorporate the phase error **436** and/or the frequency error **433** to adjust the DCO control signal **212** depends on the particular implementation of the PLL. When the PLL is locked the difference between successive measurements on the same edge of the sample clock signal is zero and therefore the frequency information from the ripple counter does not affect the

use of the much finer phase information from the TDC to maintain lock. For fractional-N operations, there is a chance the difference in the ripple counter samples will be ± 1 , and hence to avoid false triggers of phase/frequency error, embodiments only consider successive ± 1 s, or three successive ± 1 s for some embodiments. For wider range multi-modulus dividers, where more false triggers can occur, a wider range of possible false triggers are ignored based on the range of the multi-modulus divider used to achieve the desired frequency in the PLL.

[0026] In addition to the ripple counter, the feedback divider includes logic to ensure the feedback divider **222** supplies a desired feedback clock signal **208**, e.g., with a desired duty cycle, e.g., 50%. FIG. **6** illustrates an example of logic **600** included in an embodiment of the feedback divider to provide the desired feedback clock signal. In an embodiment the logic **600** guarantees a duty cycle closest to 50% rounded to the closest integer after dividing the divide value by 2. For all even divide values that will be 50% but for odd integer values it will be $(N+1)/2N$ or $(N-1)/2N$, where N is the divide value. For example, a divide by 3 yields a 66% or 33% duty cycle depending on rising/falling edge taken. The logic **600** includes set reset (SR) logic **602**, which helps determine the rising and falling edge of the feedback clock **208**. Rising edge logic **604** establishes the rising edge of the feedback clock signal. For example, the rising edge of the feedback pulse may occur when the ripple counter value is zero and thus the rising edge logic **604** may be implemented as a simple NOR gate with inputs Q0:Q4. The output **605** of rising edge logic **604** supplies the D input of flip-flop **606**, which is clocked by the DCO_OUT clock signal **216** to align the feedback clock signal with DCO_OUT. The falling edge count value logic **608** asserts when the count value of the ripple counter corresponds to the falling edge. For example, in an embodiment the output signal **609** asserts when the count value is halfway to provide a 50% duty cycle feedback clock signal. Falling edge logic **608** may be implemented as a NOR gate with appropriate inputs inverted to correspond to the desired count value. The output **609** of falling edge logic **608** supplies the D input of flip-flop **610**, which is clocked by the DCO_OUT clock signal **216** to align the feedback clock signal with DCO_OUT. Of course, other embodiments may provide other duty cycles. In addition, the rising edge may assert on a non-zero count value, e.g., in an implementation that uses a ripple counter as an up counter or that counts down to a desired value other than zero. Regardless of the particular implementation, the count values are used to determine the rising and falling edge of the feedback clock signal.

[0027] Referring again to FIG. **5**, in addition to frequency information, the value of the ripple counter provides phase information. Instead of requiring a difference between successive measurements, a single sampled value can be used to provide the phase information. For example, assume that if the reference clock (f_{ref}) and the feedback clock (fb_clk) are aligned, the sampled value is “n”. Assume the feedback divider is a divide by **24**, and the sample clock is an inverted version of the reference clock f_{ref} delayed by two DCO_OUT cycles and the sample is made on rising edge of retimed f_{ref_n} . With those assumptions, the value of the div_cnt sample for a locked PLL should be 10. If the phase is off by one DCO_OUT cycle the count will be 11 or 9 when sampled. Those count values correspond to the phase being off by a DCO_OUT cycle. Of course, other embodiments may sample at a different location but the count value at that location should indicate a locked PLL. The TDC provides much greater phase resolution than 1 DCO cycle but its range, as stated earlier, is limited. If a phase difference of several DCO cycles exists, the control signal is adjusted by a correction amount corresponding to the number of cycles. That correction amount can be stored, e.g., in non-volatile memory and correspond to the sampled count, which is an integer between 0 and 23 depending on the sampling edges and the retimed f_{ref} signal.

[0028] Referring back to FIG. **4A**, in the illustrated embodiment a delta sigma modulator **418** controls the feedback divider. The delta sigma modulator receives the desired frequency (f_{req}) **420** and the divide value of the ripple counter changes in accordance with divide control signal **422** supplied by the delta sigma modulator. The change in the divide value of the ripple counter should occur to ensure there are no unexpected perturbations in the feedback clock signal. For example,

the new divide value should be “ready” in an embodiment before the count reaches the end of the current divide (indicated by a count value of 0 in FIG. 5) so the new count value can be loaded on reset. More generally, the new divide value should be “ready” any time before the next cycle where the feedback divider has to count up/down to/from the new value.

[0029] In embodiments, the divider counts on the negative edge of the DCO clock and the sampling occurs on the positive edge of the retimed fref clock signal fref_re_n. Various timing schemes can be utilized depending on the particular implementation and hardware capabilities. The timing should be synchronized to ensure that the phase information from the TDC **202** and the phase and frequency correction information **403** based on the ripple counter count value describes the same edge of the feedback clock signal. Thus, in the embodiment illustrated in FIG. **4A**, the correction amount **403** and the filter output signal **411** should describe the same edge of the feedback clock signal. In an embodiment, the output of the ripple counter has a delay of only one flip-flop from the DCO output to ensure the ripple counter is sampled at a good time.

[0030] While the approach described herein to use the ripple counter values to provide coarse phase information and frequency information to the loop filter is particularly useful during startup as the PLL tries to initially lock to the desired output frequency, the ripple counter values can also be used when the PLL loses lock due to, e.g., a supply voltage glitch, to more quickly bring the PLL back into lock, or when the PLL tries to lock to a different frequency. While the control logic in FIG. **4A** is shown as a separate block, the control logic may be incorporated in whole or in part into the loop filter or other control logic associated with the PLL.

[0031] FIG. **7** illustrates the system when the PLL is locked. That is, fref **206** and the fb_clk signal are aligned at **702**. The control logic samples the ripple counter at **704** and **706** and retrieves the counter value of “10”, which is the counter value sampled when the PLL is locked. Thus, the phase information from the counter indicates a locked condition and the frequency information also indicates the frequency of the feedback clock signal and the reference clock is the same since both samples of the ripple counter have the same value. Thus, the phase and frequency information from the counter does not affect the digital loop filter when the PLL is locked.

[0032] Thus, a PLL providing coarse phase/frequency information from the feedback divider has been described. The description of the invention set forth herein is illustrative and is not intended to limit the scope of the invention as set forth in the following claims. The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is to distinguish between different items in the claims and do not otherwise indicate or imply any order in time, location, or quality. For example, “first phase information,” and “second phase information,” does not indicate or imply that the first phase information occurs in time before the second phase information. Variations and modifications of the embodiments disclosed herein may be made based on the description set forth herein, without departing from the scope of the invention as set forth in the following claims.

Claims

1. A phase-locked loop (PLL) comprising: a phase detector circuit coupled to receive a reference clock signal and a feedback clock signal and generate first phase error information indicative of a phase difference between the reference clock signal and the feedback clock signal; a digital filter coupled to receive the first phase error information and generate a filter output using the first phase error information, the filter output being used to generate an oscillator control signal; an oscillator configured to generate an oscillator output signal based, at least in part, on the oscillator control signal; a feedback divider coupled to the oscillator output signal and configured to generate the feedback clock signal; and wherein the feedback divider is sampled to generate frequency error information and second phase error information used to adjust the oscillator control signal.
2. The PLL as recited in claim 1 wherein the phase detector circuit is implemented as a time to

digital converter.

3. The PLL as recited in claim 1 wherein the feedback divider comprises a ripple counter.

4. The PLL as recited in claim 3 wherein the second phase error information from the ripple counter is more coarse than the first phase error information.

5. The PLL as recited in claim 3 wherein a retimed reference clock signal is used to sample the ripple counter.

6. The PLL as recited in claim 5 wherein the second phase error information is determined based on the frequency error information.

7. The PLL as recited in claim 6 wherein the oscillator control signal is adjusted according to the second phase error information.

8. The PLL as recited in claim 3 wherein a difference between a first sampled value of the ripple counter and a second sampled value of the ripple counter is indicative of the frequency error information.

9. The PLL as recited in claim 8 wherein the oscillator control signal is adjusted based on the difference.

10. The PLL as recited in claim 3 wherein a divide value of the ripple counter is programmable.

11. The PLL as recited in claim 3 wherein the frequency error information is used to provide faster lock.

12. A method comprising: generating first phase error information indicative of a phase difference between a reference clock signal and a feedback clock signal in a phase-locked loop (PLL); receiving the first phase error information at a loop filter and generating an oscillator control signal using, at least in part, the first phase error information; receiving the oscillator control signal at an oscillator and generating an oscillator output signal; and generating a feedback divider signal in a feedback divider that is coupled to the oscillator output signal; sampling the feedback divider to generate frequency error information and second phase error information; and generating the oscillator control signal based, at least in part, on at least one of the frequency error information and the second phase error information.

13. The method as recited in claim 12 wherein the feedback divider is implemented as a ripple counter.

14. The method as recited in claim 13 further comprising retiming and negating the reference clock signal to generate a retimed reference clock signal and using the retimed reference clock signal to sample the ripple counter to generate the frequency error information and the second phase error information.

15. The method as recited in claim 14 wherein the second phase error information is determined based on successive frequency error information.

16. The method as recited in claim 15 further comprising: taking a first sample of the ripple counter and taking a second sample of the ripple counter; and determining the frequency error information according to a difference between the first sample and the second sample.

17. The method as recited in claim 16 further comprising taking the first sample of the ripple counter with a first edge of the retimed reference clock signal and taking the second sample with a second edge of the retimed reference clock signal.

18. The method as recited in claim 12 further comprising using the frequency error information to provide faster lock when the PLL is locking to the reference clock signal as compared to not using the frequency error information.

19. A phase-locked loop (PLL) comprising: a time to digital converter circuit coupled to receive a reference clock signal and a feedback clock signal and generate first phase error information indicative of a phase difference between the reference clock signal and the feedback clock signal; a digital filter coupled to receive the first phase error information and generate a filter output using the first phase error information, the filter output being used to generate an oscillator control signal; an oscillator configured to generate an oscillator output signal based, at least in part, on the

oscillator control signal; a feedback divider circuit including a ripple counter coupled to the oscillator output signal and configured to generate the feedback clock signal; and wherein samples of the ripple counter are used to generate frequency error information and second phase error information and at least one of the second phase error information and the frequency error information being used to generate the oscillator control signal.

20. The PLL as recited in claim 19 wherein the ripple counter is sampled with a retimed version of the reference clock signal.
